Sixth Sense Technology: Life Beyond Physical Senses

Sixth Sense technology, based on the interaction of the physical world with digital information, has the potential of bringing about a revolution in the field of technology and paving the way for many more great innovations. Revealing the true meaning of this technology and further explaining its various applications and implementations in the real world is the primary focus of this article

Dr Rajiv Kumar Singh holds a PhD in electronics engineering from IIT (BHU), Varanasi, and is currently employed with Institute of Engineering & Technology (IET), Lucknow as assistant professor and is also acting as associate dean (PG and research), AKTU, Lucknow

Taru Meshram is an aspiring engineer who has a keen interest in the fields of science and technology. He worked on this project for more than a year at Institute of Engineering & Technology (IET), Lucknow

Sixth Sense technology uses a wearable gesture interface to augment the physical world. It allows us to make use of hand gestures for a particular set of tasks. The wearable devices used act as a gestural interface and integrates the real world with digital information.

Sixth Sense technology has the potential to bring about a digital revolution across the globe. The fact that it can be utilised in multiple ways, like performing basic actions, locating points on a map and calling someone directly without picking up the phone, asserts its importance.

Gesture recognition is a part of augmented reality (AR), which forms one of the primary basis of Sixth Sense technology. It is aimed at interpreting human gestures through program codes and mathematical algorithms.

The first person to make a device based on this concept was Steve Mann. He developed a wearable computing device in the 1990s when he was a student at Media Lab. He implemented the concept through the means of a neck-worn projector accompanied by a camera system.

The concept was further developed by Pranav Mistry, a research associate and PhD candidate at Massachusetts Institute of Technology, who is currently head of Think Tank team and vice president of research at Samsung Research America.

Working and design of the device

Many devices have been developed based on Sixth Sense technology. One such example is Wear Ur World (WUW) device. This prototype device gathers data from the user's environment, answers questions and fulfills tasks using the Internet as an information bank and then presents the result back to the user by means of a display mechanism.

Simple computer-based vision-oriented codes and gesture identification and interpretation technologies are used to evaluate inputs received from the individual using the device in the form of images or gestures produced by the user's hands or face. Application of this device includes every prominent

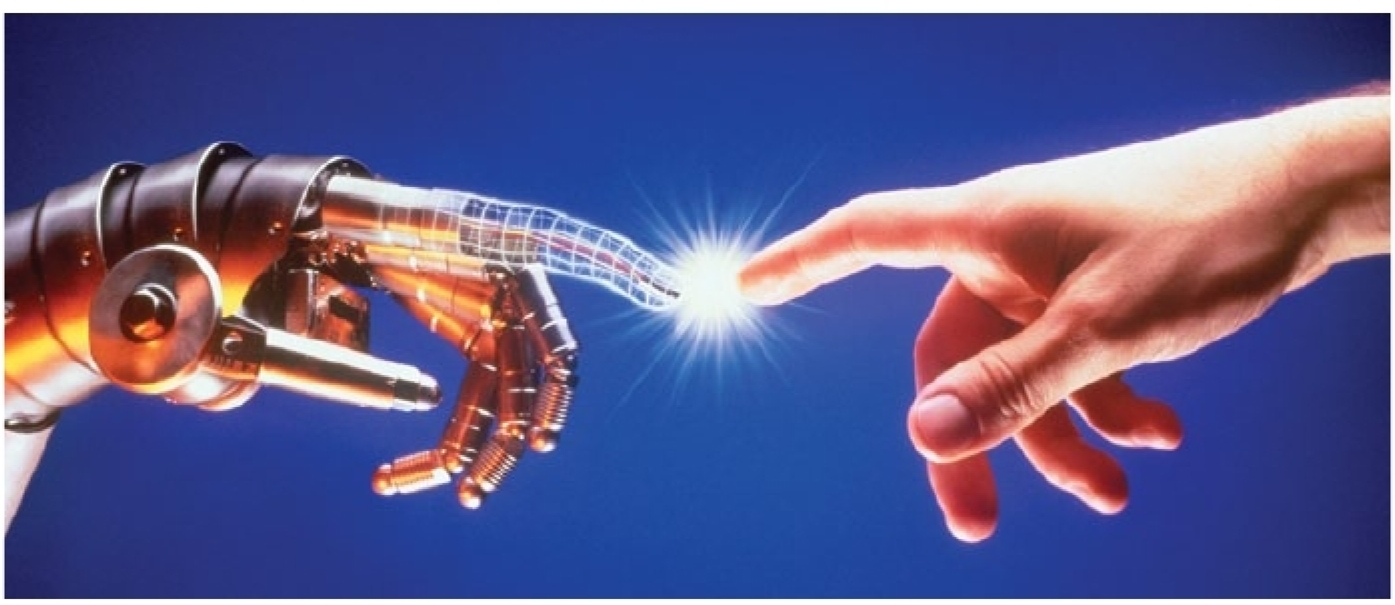

Fig. 1: Interaction of the physical world with the digital world (Credit: http://tecnologymedia-com)